

AIR Guitar: A Browser-Native Number-Gesture Instrument for Everyday Music-Making

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ABSTRACT AIR Guitar is a pervasive, browser-native number-gesture instrument by eliminating the specialized hardware and installation friction that typically impede music technology adoption for older adults. Running entirely in the web browser, it achieves real-time performance. For gesture recognition, it uses MediaPipe keypoints with a lightweight client-side Support Vector Machine (SVM) classifier. This system is key to pervasive deployment: the SVM achieved comparable chord accuracy (~96.1%) to a compact CNN while incurring a significantly lower frame-rate drop (33% vs. 63%), a critical trade-off for responsive use. For inclusivity, it centers on motor-friendly number gestures (1–7), mapped via the Nashville Number System for easy transposition, which reached up to 99.6% accuracy. Gestures are rendered with Web Audio/MIDI and integrated with a repertoire workflow supporting culturally familiar MIDI for sing-along. This zero-setup system was successfully deployed with older adults in Taiwan's Community Care Stations, demonstrating its real-world efficacy.

Index Terms— pervasive computing, accessible computing, human–computer interaction (HCI), browser-based systems, gesture recognition, older adults, music technology, Web Audio/MIDI.

INTRODUCTION

Pervasive computing aims to integrate technology seamlessly into daily life, supporting areas like later-life well-being and social connection. Yet, music technology for older adults often fails to be truly pervasive, blocked by persistent barriers: specialized hardware, complex installation, steep learning curves, and limited culturally familiar repertoire [1]. This deployment friction directly undermines pervasiveness. Our primary goal with AIR Guitar is simple: develop a system that runs instantly and seamlessly on devices people already own, effectively removing this friction.

We introduce AIR Guitar, an assistive, inclusive, real-time instrument. While previous work broadened participation—such as Mathews's Radio Baton [2] or head-motion instruments [3]—these systems often relied on purpose-built sensors or complex setups that are hard to sustain outside of labs. Our approach is fundamentally web-native. Using only a webcam, the entire instrument stack runs in the browser: Google MediaPipe extracts real-time hand keypoints, a client-side Support Vector Machine (SVM) classifies gestures, and Web Audio/MIDI handles low-latency sound rendering. This complete, browser-based pipeline ensures the tool is practical and deployable on ordinary laptops in community settings.

Building on accessible music education principles [15], AIR Guitar centers its Human-Computer Interaction (HCI) on the number-gesture interface, aligning with the Nashville Number System. Simple hand poses (1–7) map to scale degrees, simplifying complex harmonies. This design proved essential for inclusivity: number gestures were highly approachable, achieving up to 99.6% accuracy and contrasting favorably with complex chord shapes that challenged older adults. The zero-setup system, which includes a shareable URL for personalized settings and repertoire, was successfully deployed in Taiwan's Community Care Stations, supporting social sing-alongs and demonstrating real-world efficacy. The remainder of the paper details the system design,

implementation trade-offs, and experimental validation.

This paper presents the following key contributions to the fields of pervasive and accessible computing:

1. **Pervasive System Architecture:** Presenting an end-to-end, browser-native instrument that runs entirely within the client's web browser. This realization of the system architecture allows for a zero-installation, ubiquitous deployment model on commodity devices.
2. **Efficiency Analysis for Pervasiveness:** Providing an empirical validation of the lightweight MediaPipe-SVM pipeline against a heavier deep learning approach. The analysis details the performance trade-off, highlighting the pipeline's utility in minimizing frame-rate degradation (approx. 33% FPS drop) required for responsive pervasive use.
3. **Accessible HCI Design and Evaluation:** Demonstrating the design and evaluation of a motor-friendly number-gesture interface (achieving up to 99.6% accuracy). This low-cognitive-load approach, derived from the Nashville Number System, offers a valuable accessible alternative to complex chord shapes.
4. **Community-Ready Repertoire Workflow:** Introducing a shareable, personalized MIDI workflow that supports culturally relevant songs and synchronizes on-screen cues (lyrics/melody) to facilitate sustained, joyful group practice in community settings.

The demonstration video for the AIR Guitar system is available at: <https://youtu.be/ESsW2KcjUp0>.

RELATED WORK

This section reviews related work across human-computer interaction, machine learning for real-time performance, and web-native digital musical instruments, establishing the foundation and highlighting the novel distinctions of the AIR Guitar system.

Human-Computer Interaction and Accessible Gesture

Hands are a natural interface, making gesture interaction a core theme in HCI. Projects have explored expressive and accessible interfaces, such as real-time sign language translation to reduce communication barriers [4], [5], and using Kodály hand signs as gestural queries for music retrieval [6]. These systems demonstrate how carefully chosen gestures can make interaction both expressive and inclusive.

Specifically in music, digital instruments have long explored alternative performance methods. Mathews's Radio Baton let conductors shape tempo, dynamics, and expression with hand movements [2], and Davanzo and Avanzini demonstrated hands-free performance using only head motion [3], pointing to accessible designs based on unconventional sensing.

While such work broadens participation, many existing digital musical instruments still depend on purpose-built sensors, custom hardware, or complex setups that are hard to sustain outside labs or clinics. We adopt the principle of simple, carefully chosen gestures by favoring learnable number gestures over complex chord shapes, making the instrument highly accessible and inclusive for users with varying levels of motor control, such as older adults.

Machine Learning Trade-Offs for Real-Time Pervasive Systems

Deep learning (DL) dominates many image-recognition tasks, but for resource-constrained, real-time applications with modest datasets and tight latency budgets, classic models such as Support Vector Machines (SVMs) remain a highly attractive solution. Convolutional Neural Networks (CNNs) can sometimes achieve higher accuracy [7], [8], but this typically comes at the cost of higher compute, memory consumption, and longer training time. Hybrid pipelines are often used to achieve the best of both worlds, capturing skeletal features [9] (like those from OpenPose) before recognition to reduce background sensitivity and improve

robustness.

Guided by these performance trade-offs—which are critical for pervasive deployment—AIR Guitar adopts a MediaPipe-SVM design. We leverage MediaPipe's efficient, real-time hand keypoints [10] and operate a lightweight SVM on compact geometric features (segment lengths, adjacent-fingertip distances, and joint angles). This design ensures a critical balance between recognition accuracy, responsiveness (<1 ms classification latency), and the ability to run entirely in-browser on commodity devices.

Web-Native Digital Musical Instruments (DMIs)

The convergence of art and science drives the advancement of musical tools and forms of expression. Modern gesture-driven instruments extend this line of work. Examples include networked air-instrument systems for non-musicians performing collaborative music remotely [11] and MediaPipe-based models that translate finger positions into corresponding piano notes [12].

Critically, several projects have demonstrated how musical control can live entirely on the web:

1. Handmate [13] is a browser-based hand-gesture controller that utilizes Web Audio, Web MIDI, and Open Sound Control (OSC).
2. MuGeVI [14] tracks 21 hand keypoints and offers multiple performance modes, often coupled with Max/MSP for rich musical interaction.
3. Our prior work explored a web-based MIDI platform for accessible music education, specifically Pachelbel's Canon [15].

AIR Guitar builds on the web-native foundation of Handmate and MuGeVI but distinguishes itself in two key ways necessary for widespread pervasive adoption:

1. **Deployment Model:** It runs end-to-end in the browser (webcam, MediaPipe, SVM, Web Audio/MIDI) using an optimized lightweight SVM, completely eliminating the setup, installation, and maintenance overhead associated with standalone applications or server-side processing.
2. **Accessible HCI:** It centers on number gestures (1–7) based on the Nashville Number System [16][17], which dramatically reduces motor and cognitive load for older adults, providing a more accessible vocabulary than systems that rely on complex finger-bending detection.

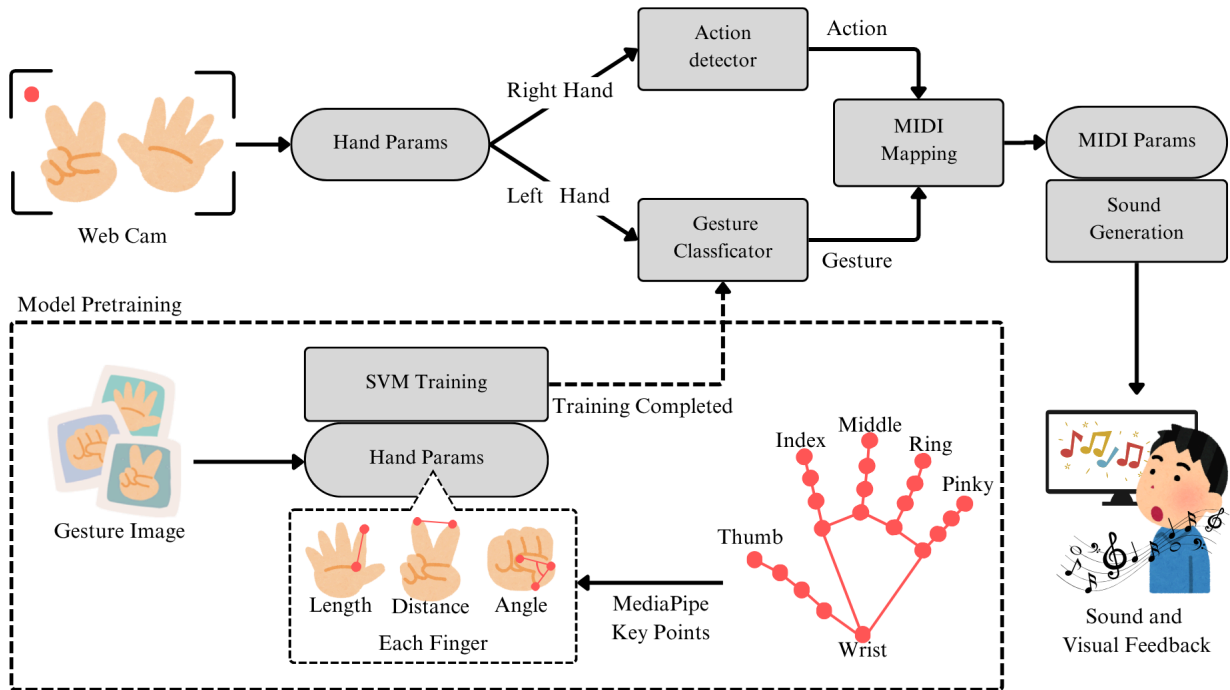


Fig. 1. Webcam video is analyzed by MediaPipe to obtain hand keypoints; compact features feed an SVM (left hand) for number/chord labels and an action detector (right hand) for strum/pluck. Signals are mapped to MIDI and rendered via Web Audio or Web MIDI with on-screen cues. The lower panel shows offline feature extraction and SVM training.

Design Framework

The AIR Guitar architecture is designed for browser-native, pervasive deployment, prioritizing low-latency real-time performance on commodity hardware with zero installation friction. Fig. 1 shows the end-to-end, in-browser pipeline. A webcam streams video, providing a mirrored selfie view so on-screen motion matches the user’s perspective. The video is processed by MediaPipe Hands, which returns a 21-keypoint skeleton per hand per frame.

We divide the process into two parallel streams for musical control: the Right Hand triggers musical actions (strum/pluck), and the Left Hand forms chord or number gestures. For accessibility, the roles of the hands can be swapped at runtime. Signals converge in a MIDI-mapping stage that derives voicing, velocity, and timing. Sound is rendered via Web Audio or routed through Web MIDI to an external synthesizer. The entire model is trained offline and shipped with the web application, loading directly in the browser with no server dependency.

Augmented Features before Pretraining

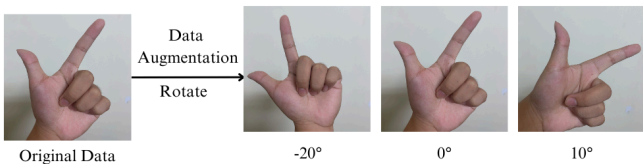


Fig. 2. Number 7 (Bdim Chord) Gesture and Data Augmentation

We built a lightweight, robust classifier by carefully engineering the feature space and augmenting the training data. The dataset was created from the 7 basic chords on the C major scale—C, Dm, Em, F, G, Am, Bdim—which also define

the number gestures (1–7). For each class, 40 raw examples were recorded and then augmented by in-plane rotation (-20° , 0° , $+10^\circ$) to create 120 samples per class, yielding 840 total samples per vocabulary. This augmentation, shown in Fig. 2, improves robustness to camera pose and hand orientation, which is crucial for reliable performance in unconstrained, real-world settings.

From the 21 MediaPipe keypoints, we compute a compact feature vector that encodes hand shape. This vector consists of geometric features: lengths (phalange segments), adjacent-fingertip distances (e.g., thumb–index), and joint angles. These geometric features are more robust to background and camera variation than raw pixels. We apply proportional normalization (e.g., by palm width or wrist–index span) to reduce the effect of hand-size and camera-distance on the features, creating stable inputs for the SVM.

Number Gesture-to-Chord MIDI mapping

Verse	Am	F	C	G	
Lyrics:	we keep this love in a photograph we made these memories for ourselves				
Roman numerals:					
C = 1 st	vi	IV	I	V	
Nashville Number:					
C = 1 st	6-	4	1	5	

Fig. 3. Number-gesture for the song “Photograph”.

Fig. 3 illustrates how the Nashville Number System is used to map chord progressions to simple hand gestures, allowing for easy key transposition. This section details how these number gestures are translated into MIDI output.

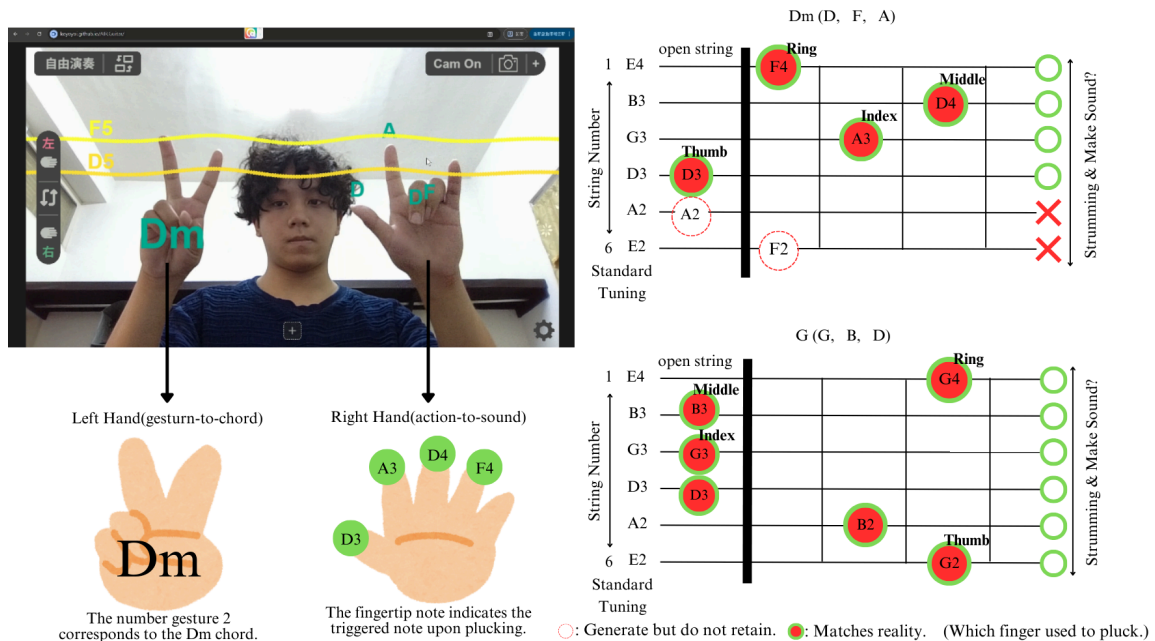


Fig. 4. Gesture-to-chord mapping and voicing generation. The runtime uses a selfie (mirrored) camera view so on-screen motion matches the user’s perspective; the left hand selects a chord/number, and the right hand plucks/strums. Visual cues confirm the detected gesture and trigger.

The left hand switches chords using number gestures; for example, number gesture “2” corresponds to a Dm chord (D, F, A). The system first maps the chord tones onto the six strings of a standard-tuned guitar (E2, A2, D3, G3, B3, E4), producing an initial layout (Fig. 4 shows an example for Dm and G chords). To create a realistic guitar voicing, the system then removes notes below the lowest root note, resulting in an authentic and playable sound using a standard MIDI sound-font. For example, the Dm chord voicing is trimmed to D3, A3, D4, F4.

The Right Hand is dedicated to musical actions. Strumming is recognized by tracking wrist, elbow, and shoulder key points, enabling detection of strumming direction (up or down) and velocity based on arm angle changes. For plucking, the system monitors finger joint bending to determine which string should be triggered, and the note velocity is adjusted proportionally to the bending angle. Both strumming and plucking events are encoded as MIDI signals [18] and forwarded to the synthesizer. Finally, the system includes a capo adjustment function: raising either hand changes the capo position, which allows seamless key transposition while preserving the finger positions and chord structures.

Shareable, Personalized MIDI Repertoire

AIR Guitar is built around the MIDI file format so songs can be loaded, practiced, customized, and shared with minimal setup. Users upload their own MIDI files or choose from a curated repertoire of culturally relevant classics. On load, the system parses the MIDI, converts harmony to Nashville-style numbers for instant transposition, and extracts melody for on-screen prompts. Lyric meta events (FF 05) are synchronized to provide a karaoke-style display that integrates chords, melody, and words.

A core feature for pervasive, unadministered use is Personalization: key, tempo, hand modes, font size, and display options are fully customizable. These settings, along with the song choice, are encoded in a shareable URL or QR code. This allows facilitators to prepare playlists without needing user accounts or installation, and recipients can open the link and get the exact same personalized view instantly on any webcam-equipped device. The platform offers built-in Web Audio for sound immediately, or can route events via Web MIDI for advanced users with external synthesizers.

Summary of Pervasive Design: The entire Design Framework is governed by the principle of zero-friction deployment. By using the browser for video input, lightweight client-side machine learning for real-time classification, and self-contained repertoire sharing via URLs, AIR Guitar overcomes the common hardware and software barriers to technology adoption in community and long-term care settings.

Experimental Results

The experimental validation of AIR Guitar focuses on two critical aspects: the accuracy and friendliness of the accessible Human-Computer Interaction (HCI) design and the efficiency and real-time performance trade-offs of the machine learning pipeline, which are essential for pervasive deployment.

Gesture Vocabulary Performance

We evaluated the performance of two distinct hand gesture vocabularies: complex chord shapes (C, Dm, Em, F, G, Am, Bdim) and motor-friendly number gestures (1–7). The data from the Support Vector Machine (SVM) training confusion matrices confirms the superiority of the numeric vocabulary for inclusive, real-time use.

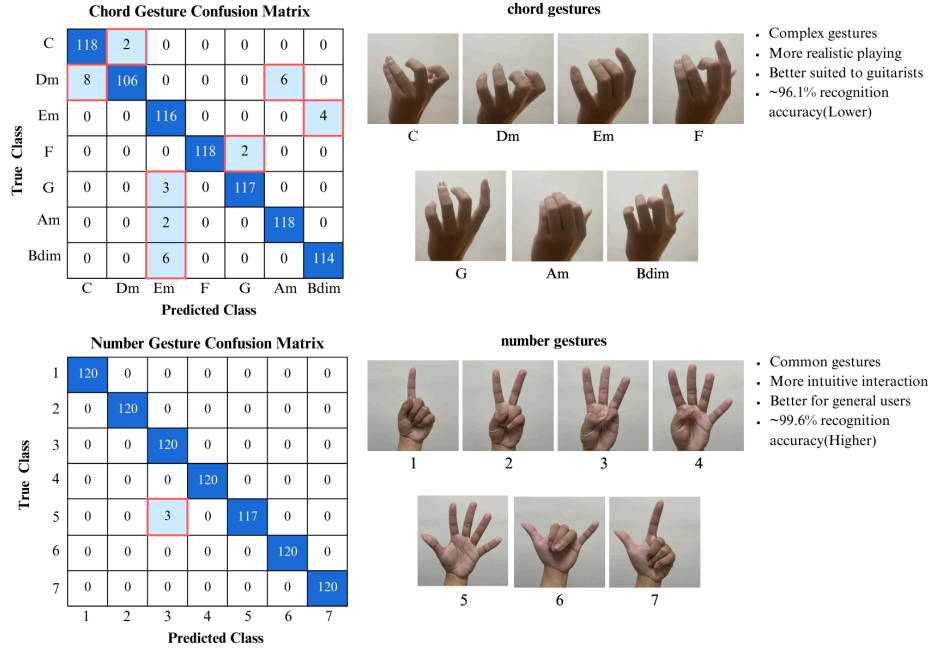


Fig. 5. The SVM training confusion matrices and comparison between gestures of chords and numbers. Red boxes indicate classes that are prone to misclassification.

Number Gestures (HCI for Accessibility): The simple number gestures proved highly effective and motor-friendly, achieving an exceptional recognition accuracy of 99.6% (837/840 samples). The only noted recurring misclassification was a minor slip from 5 to 3 (3 cases), primarily occurring when the user under-extended the ring finger. This result validates the use of everyday counting poses as a low-cognitive-load alternative for older adults.

Chord Gestures (Traditional Performance): The more complex chord shapes achieved an accuracy of 96.1% (807/840 samples). The denser off-diagonals in the confusion matrix reflect the higher motor and visual pressure of these gestures, especially for novices. The most frequent errors were: Dm recognized as C (8 cases), Dm as Am (6 cases), Bdim as Em (6 cases).

This outcome guided the system design to default to number gestures for primary interaction and offer chords as an optional mode, enhancing the system's overall accessibility.

Model Comparison: Efficiency for Pervasive Systems

Pervasive computing demands systems that run efficiently on resource-constrained commodity devices. We compared the lightweight MediaPipe-SVM pipeline against a compact three-layer Convolutional Neural Network (CNN) in the same Python codebase to validate this critical trade-off.

Frame-Rate Reduction (FPS Decrease): The CNN, despite a marginally higher chord accuracy (97% vs. 96.1%), incurred a 63.3% frame-rate drop. In contrast, the SVM achieved comparable accuracy with only a 33.3% frame-rate drop. This substantial difference in computational overhead is the key persuasive point, demonstrating the practical utility of the MediaPipe-SVM pipeline for maintaining a responsive user

experience on ordinary, non-specialized hardware.

Table I. SVM and CNN Comparison

	<i>SVM (chord)</i>	<i>SVM (number)</i>	<i>CNN (chord)</i>
Recognition Accuracy	~96.1%	~99.6%	~97%
FPS Decrease	~33.3%		~63.3%
Gesture Latency	< 1 ms		
Sound Latency	Plucking ≈ 0 ms, Strumming < 100ms		
Training Time	Fast		Longer

Low Latency: The SVM's classification latency is less than 1 ms, ensuring the system provides a real-time, gesture-driven musical experience. The total sound latency is also low: 0 ms for plucking and < 100ms for strumming.

Adaptability: The SVM's fast training time is also an advantage, enabling quick, per-user adaptation to different hand sizes or motor capabilities, which is valuable for inclusive deployment.

Accuracy alone favors the numeric vocabulary; friendliness amplifies that advantage. Number gestures rely on everyday counting poses, tolerate partial flexion, and remain distinctive under modest camera changes, which shortens onboarding and reduces fatigue. Chord shapes mirror real fingerings and are pedagogically useful, but several demand simultaneous flexion and abduction that many novices and older adults find difficult to hold. The denser off-diagonals for chords in Fig. 5 reflect these motor and visual pressures. Design response: default to numbers for primary interaction, expose chords as an optional mode, and add light temporal smoothing plus targeted augmentation around the Dm/Bdim neighborhood to reduce slips.



Fig. 6. An elderly woman (left) and an elderly man (right) in a long-term care community in Taiwan performed the beloved Taiwanese song *Longing for the Spring Breeze* using a sequence of small pinches or waving the hands. A video is available at: <https://youtu.be/0Xur92Akb8k>

Deployment on the Web and Pervasive Accessibility

AIR Guitar's core philosophy is to eliminate friction and ensure broad accessibility. We migrated our Python prototype to a browser-native JavaScript application, enabling pervasive deployment in non-technical settings like community centers and homes.

Zero-Setup: The system runs instantly in any standard web browser on commodity devices. By requiring only a webcam and zero installation, we successfully bypass the setup barriers that commonly restrict the adoption of music technology.

Audio Flexibility: The platform defaults to built-in Web Audio for immediate feedback, while Web MIDI remains an option for advanced users seeking external synthesis.

System Availability: AIR Guitar is publicly available at: <https://imuse.ncnu.edu.tw/AIR.Guitar>.

The AIR Guitar system is designed not just as a guitar simulator, but as a powerful, versatile virtual instrument. Through the use of appropriate soundfonts or external synthesizers via Web MIDI, the system's output is not limited to guitar sounds.

Here is a specific example showcasing that we played the melody of the song *Longing for the Spring Breeze* (using a single-handed, expressive mode) is available here: https://youtu.be/LIVDYFkh_rs. As demonstrated in the video, the user's right hand easily triggers melody notes (note-on/note-off) via a simple pinch and release gesture. This action frees the left hand for expressive control: its vertical position (height) is mapped to MIDI velocity (loudness), and lateral vibration or magnitude of movement controls musical details such as vibrato.

While achieving this level of expression requires a period of practice, even for younger users, this design choice makes advanced musical performance more accessible and intuitive than traditional controllers. This approach, which separates rhythmic triggering from expressive control, is critical for

making nuanced music performance friendlier for all users, including older adults.

Applications in Long-Term Care Centers

AIR Guitar was successfully deployed in older-adult programs within Taiwan's Community Care Stations. This application supports group music and social sing-along, demonstrating the system's real-world efficacy.

Figure 6 shows an older adult using AIR Guitar with a simple Single-Handed Performance Mode. To make the system more accessible for elderly participants, a single-handed performance mode was extended and modified. In this mode, when a MIDI file is loaded:

1. The system provides real-time chord or note prompts on the screen (similar to a karaoke-style display).
2. It plays according to the MIDI score.
3. Users interact using only hand gestures or movements.

Reduced Cognitive Load: This design aims to make the interaction process more intuitive and seamless while reducing the cognitive load during operation.

Participant Feedback

Feedback from 18 older adults in two care centers was instrumental in confirming the system's inclusive design choices.

Preference for Simplicity: Participants generally found the system enjoyable and engaging. However, they preferred single-hand operation over two-hand control and noted that fine mid-air gestures could be challenging to control given variability in hand mobility.

Effective Design Elements: In contrast, larger strumming motions felt natural and easy to perform. The number cues aligned with lyrics in the score display were consistently cited as being helpful for both operation and comprehension. Most participants reported a willingness to continue practicing.

CONCLUSION

We presented AIR Guitar, a system built to fulfill the promise of pervasive computing by delivering an accessible digital instrument on commodity, browser-native devices. Our architecture successfully eliminates the hardware and installation friction that typically impedes music technology adoption in everyday environments like community care centers. The core of AIR Guitar is a lightweight, real-time system and a design focused on accessibility:

1. **Pervasive System Design:** The use of MediaPipe hand keypoints with a lightweight SVM classifier enables end-to-end, in-browser operation with very low latency (<1 ms classification). Compared to a compact CNN, the SVM achieved comparable chord accuracy (96.1% vs. 97%) while incurring a significantly lower frame-rate drop (33% vs. 63%), a critical trade-off for responsive deployment on ordinary laptops.
2. **Accessible HCI and Community Impact:** The design centers on simple number gestures (1–7) mapped via the Nashville Number System, proving more motor-friendly and approachable for older adults than traditional chord shapes. The gesture vocabulary reached up to 99.6% accuracy in testing, which is amplified by a repertoire workflow that supports importing culturally familiar MIDI with synced lyrics for practice and group sing-alongs. Community sessions demonstrated that the system is quick to learn and effective for social music participation.

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